

Figure 1.3 Albert Einstein.

The name of this twenty-five-year-old, obviously, is Albert Einstein.

How does he do it? The idea is surprisingly simple. Anyone could have arrived at it, from the time of Democritus onwards, if he had had Einstein's acumen and a sufficient mastery of mathematics to make what was not an easy calculation. The idea goes like this: if we observe attentively very small particles, such as a speck of dust or a grain of pollen, suspended in still air or in a liquid, we see them tremble and dance. Pushed by this trembling, they move, randomly zigzagging, and so they drift slowly, gradually moving away from their starting point. This motion of particles in a fluid is called Brownian motion, after Robert Brown, a biologist who described it in detail in the nineteenth century. The typical trajectory of a particle dancing in this manner is illustrated in figure 1.4. It is as if the small particle is receiving blows randomly from each side of it. In fact, it isn't 'as if' it were being hit, it really is hit. It trembles because it is hit by individual molecules of air, which collide with the particle at times from the right and at times from the left.